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DemandIQ

**Demand Planning, IBP & New Product Launch
Decision System for Premium Outdoor Apparel**

From stockout-adjusted forecasting and weekly S&OE decisions to cold-start launch planning, rolling forecasts, and lifecycle handoff.

DEMAND PLANNING | FORECASTING | IBP | S&OE | NEW PRODUCT
PLANNING

Portfolio simulation · PUBLIC / SYNTHETIC / DERIVED data · not built from any real company's internal data.

DemandIQ at a Glance

Everything you need to read the rest of this document.

A · PRODUCTS

MATURE · OWN HISTORY

APS-001 Alpine Performance Shell

CTS-001 Core Technical Shell

IMH-001 Insulated Midlayer Hoody

NEW · NO OWN HISTORY

HIS-001 Hybrid Insulated Shell

HIS-001 has no own demand history at launch, so it follows a cold-start / analog-led process rather than the mature forecasting engine.

B · CHANNELS

ECOM

Direct online

RETAIL

Own stores

WHOLESALE

Partner

D · WHAT IS DEMANDIQ?

A **synthetic premium-outdoor-apparel demand-planning simulation**. It moves from retail demand signals through demand reconstruction, forecasting, scenario planning, inventory/service risk and governed planner actions — then extends the same environment to a **new-product launch**: analog selection, an 18-month cold-start forecast, IBP consensus, initial buy, sell-through, reforecasting, FVA and rolling updates. Data is **PUBLIC / SYNTHETIC / DERIVED**; not Arc'teryx data; economics are **planning exposure proxies**.

C · ESSENTIAL PLANNING TERMS

WAPE

Weighted Absolute % Error — accuracy (lower = better)

Bias

Whether forecasts run above or below demand

IBP

Integrated Business Planning — the consensus plan

S&OE

Sales & Operations Execution — weekly execution

FVA

Forecast Value Add — did a step improve accuracy?

V0→V3

Analytical → Commercial → Consensus → Approved

WOS

Weeks of Supply — how long inventory lasts

Fill Rate

Share of demand inventory can serve

Cold Start

Forecasting a product with little/no own history

THE CASE

The Planning Challenge — and What DemandIQ Found

THE PLANNING CHALLENGE — WHY IT ISN'T SIMPLY "PREDICT SALES"

- **Sales can be censored by stockouts** — observed sales understate true demand.
- **Weather creates demand uncertainty** — but future weather can't be used at forecast time.
- **Aggregate fill can hide weekly service failures.**
- **Forecasts must become inventory decisions** — a number is not an action.
- **New products have no own history.**
- **Plans must learn as evidence arrives** — governed reforecasting, not ad-hoc overrides.

EXECUTIVE FINDINGS

MATURE 13-WK BASE DEMAND

36,435

units (portfolio)

PORTFOLIO BASE FILL

98.77%

vs 92% target

WEEKLY-SERVICE P1 SERIES

3 of 9

ESCALATE

HIS-001 V3 APPROVED 13-WK

~7,577

unconstrained demand

HIS-001 BUY · LAUNCH FILL

~8,259

units · ~97.7% fill

CYCLE-02 LIKE-FOR-LIKE

-3.05%

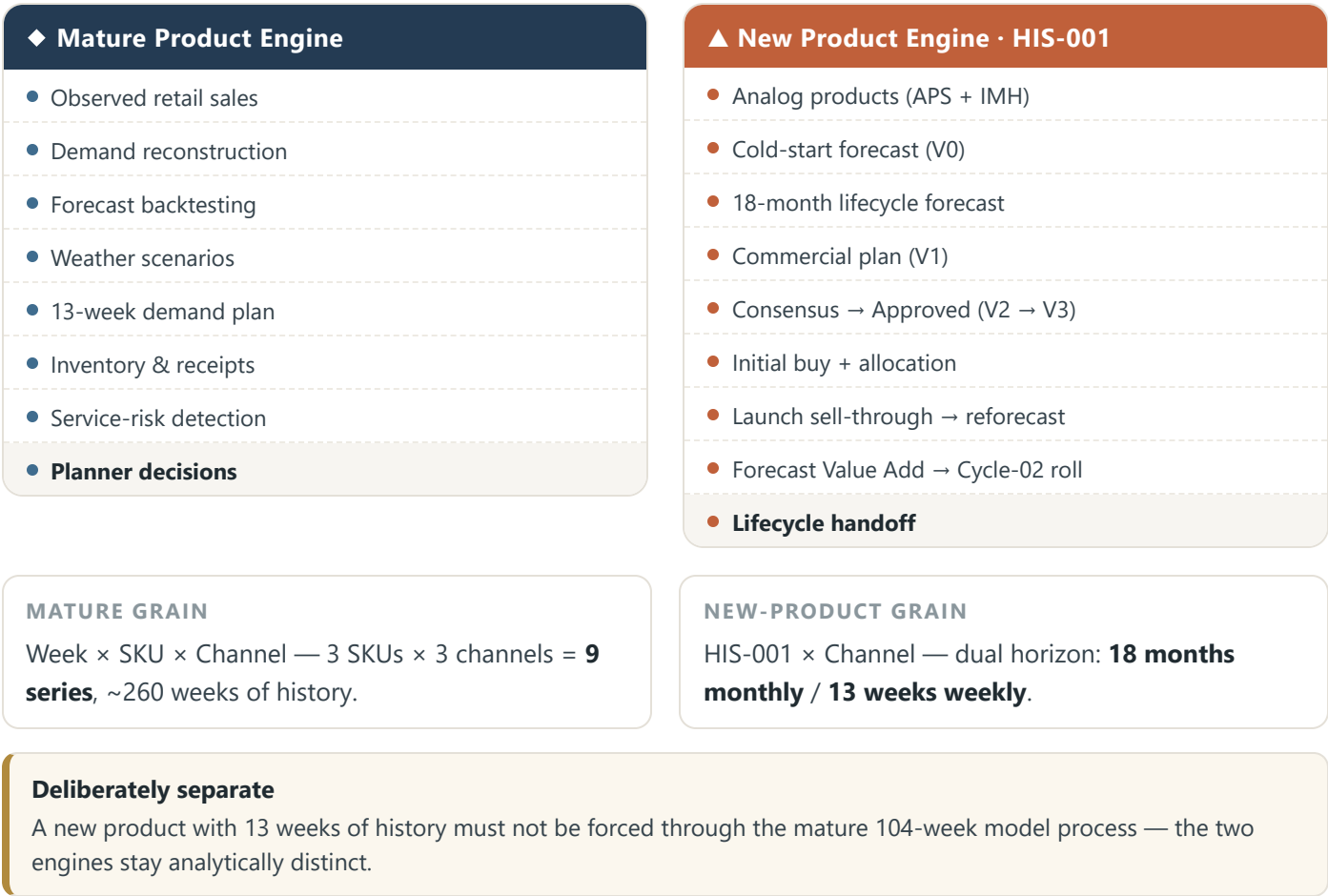
rolling revision

The through-line

Aggregate forecast accuracy and aggregate fill are **necessary but not sufficient**. The value is in the **decision** — which weeks and channels are at risk, what to buy for a product with no history, and how the plan should learn after launch.

End-to-End DemandIQ Planning Architecture

Two governed engines share one planning environment.



Recover the Demand Signal, Then Forecast It Honestly

A Why sales ≠ demand

When inventory constrains availability, sales are **censored** — forecasting them directly teaches the model that stockout weeks were low-demand weeks and biases the plan downward. DemandIQ **reconstructs demand before forecasting**, selecting **Seasonal Profile Imputation** under one governed evaluation.

CENSORED-ROW WAPE

26.08%

BIAS

-2.61%

LOST-DEMAND RECOVERY

95.68%

FULL-STOCKOUT RECOVERY

98.43%

Forecast target = **reconstructed_demand_units**. Hidden **true_demand_units** scores the reconstruction only — never a forecast feature.

Defensible over sophisticated

Where SARIMA beat ETS by only ~0.06 pp WAPE, the simpler ETS was kept; where a moving-average baseline won outright, it was kept. Selection was driven by repeated out-of-sample performance, bias and simplicity — not complexity.

B Mature forecast engine

All 9 series run an identical, honest backtest:

Seasonality	52 weeks
Initial training	104 weeks
Forecast horizon	13 weeks
Expanding folds	12
Primary metric	WAPE (bias tracked)

CHAMPION MIX

8 ETS

+ 1 baseline · 0 SARIMA

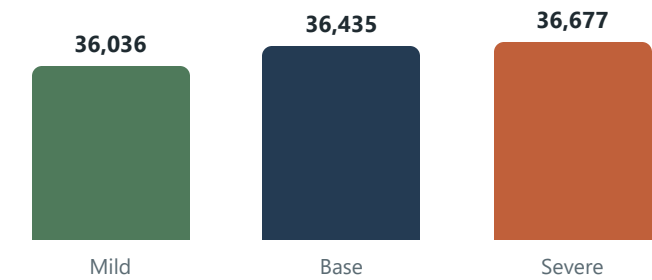
CHAMPION WAPE

8.4–21.7%

bias -3.1% to +2.4%

From 13-Week Outlook to Planner Decisions

A · WEATHER & DEMAND OUTLOOK



Weeks 1–3 **NOWCAST_REQUIRED** (none supplied → no near-term weather invented); Weeks 4–13 Mild/Normal/Severe. **No realized future weather.** Band ~1.8% of Base — weather is not the dominant near-term risk.

B · SERVICE & INVENTORY RISK

PORTFOLIO BASE FILL		SERVICE TARGET	
98.77%		92%	
P1 SERIES (ESCALATE)	13W	WORST WK	
APS-001 · WHOLESALE	93.5%	33.2%	
CTS-001 · RETAIL	95.6%	65.4%	
IMH-001 · WHOLESALE	96.1%	74.1%	

Aggregate health hides weekly risk
98.77% portfolio fill looks solved — yet **3 of 9 series** miss the weekly target twice, sharing a worst week of **2026-08-24** where committed receipts are **zero across all nine series**. Only thin-buffer channels break (not proven causation).

3 ·
ESCALATE
weekly-service P1

6 · PROTECT
low-coverage P2 · all 9 below 2.5-wk policy

Risk detection ≠ execution authorization
0 auto-chase · 0 reallocation — lead-time/transfer feasibility isn't modeled. Base exposure ≈ **CAD 192K**
PLANNING EXPOSURE PROXY, ~64% CTS-001/RETAIL — used only to rank the queue.

HIS-001 — Cold Start & Dual-Horizon Forecast

NEW PRODUCT Hybrid Insulated Shell · launched 31 Aug 2026 · **no own demand history** → the mature engine can't be applied.

A Analog selection

CANDIDATE	SCORE	ROLE
APS-001	0.679	Primary · 60%
CTS-001	0.531	Excluded
IMH-001	0.448	Secondary · 40%

APS anchors shell, price, scale & Fall/Winter seasonality. IMH uniquely supplies **insulation** APS lacks (FW correlation 0.99). CTS ranked #2 but is a redundant shell filling no gap → excluded. Blend: **60% APS + 40% IMH**.

B V0 cold-start forecast

13-WEEK V0 ~ 6,888	FIRST-YEAR V0 ~ 27,480
18-MONTH V0 ~ 44,919	SECOND-SEASON FACTOR 1.00 no pre-evidence growth

Launch-scale factor 0.60 on the blended analog. Channel mix 45 / 35 / 20.

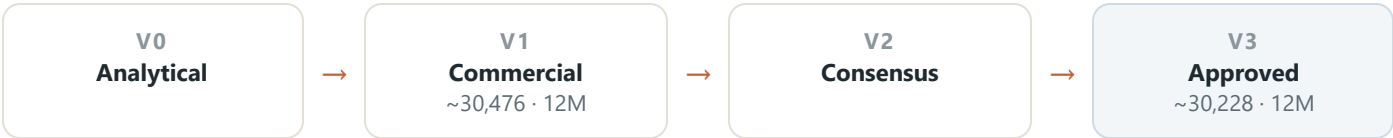
C Why two horizons

STRATEGIC · IBP
18 months, monthly — commercial, category and supply discussions.

OPERATIONAL · S&OE
13 weeks, weekly — near-term launch execution and the initial buy.

IBP Consensus → V3 → Initial Buy

A · FORECAST VERSION JOURNEY



V3 = APPROVED UNCONSTRAINED DEMAND ($V3 / V0 \approx 1.10$) — first 12 months ~30,228 · 18 months ~49,411. It is a demand position, **not** a supply-constrained plan.

B · INITIAL BUY

Approved V3 13-week demand	~7,577
+ Launch uncertainty buffer (9%)	~682
= Frozen initial buy (BALANCED)	~8,259
Flex reserve (held)	~991

The 9% launch buffer is a **launch-specific uncertainty buffer** — **not** the mature 2.5-week safety-stock rule.

C · CHANNEL ALLOCATION



SYNTHETIC SUPPLY PLANNING ASSUMPTIONS

~8-week replenishment lead · chase $\leq 15\%$ of the buy.

Launch Execution & Reforecast

A · SEEDED LAUNCH RESULT SYNTHETIC SEEDED LAUNCH SIMULATION · SEED 7

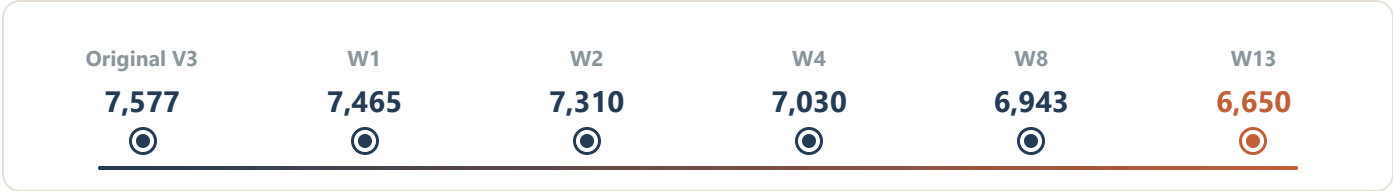
OBSERVED SALES
~6,650

LATENT DEMAND · EVAL ONLY
~6,809

LAUNCH FILL
~97.7%

The **latent demand is evaluation-only** — the operational planner never had access to hidden demand truth, and it never entered any reforecast or decision.

B · REFORECAST JOURNEY — PLANNER LEARNING OVER TIME



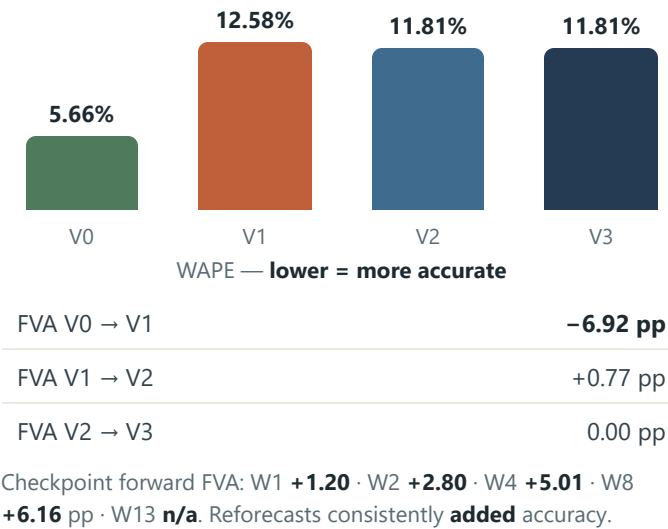
As launch evidence accumulated, the plan **progressively reduced its reliance on the pre-launch analog prior** (weighting $w = n / (n + 4)$). The planning point matters more than the formula: actual evidence gradually earns more weight.

C · A disciplined non-action

Every checkpoint resolved to **HOLD** — attainment stayed within tolerance, channel-mix deviation stayed under the reallocation trigger, coverage stayed adequate. No CHASE/CUT/REALLOCATE was *forced*. Disciplined governance, not a missing feature.

Forecast Value Add & Channel Learning

A · DID EACH INTERVENTION IMPROVE ACCURACY?



B · CHANNEL LEARNING (OBSERVED / OPERATIONAL)

OBSERVED ECOM MIX

~49.2%

vs 45% planned

ECOM LOST DEMAND

~159

W13 stockout

IDLE FLEX RESERVE

~991

REALLOCATE TRIGGER

8 pp

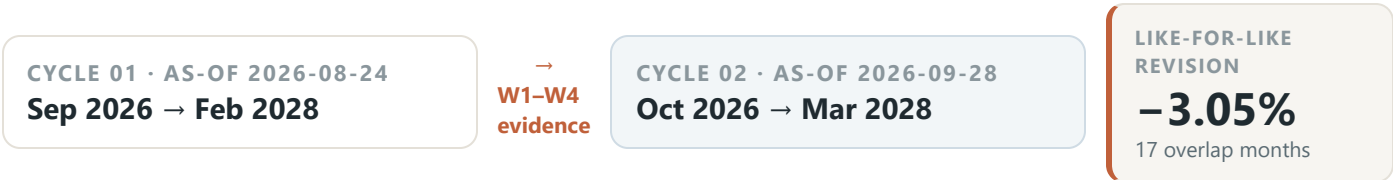
Different planning problems

Inventory remained available in reserve and other channels while ECOM stocked out — a **channel-allocation / policy-timing** issue, surfaced as a next-cycle learning. No alternative threshold is claimed optimal.

⚠ **Mandatory caveat.** The synthetic launch generator was **centered on the V0 analytical baseline**. So V0 out-scoring V1/V2/V3 on this seeded path is an **illustrative FVA governance demonstration** — **not independent empirical evidence that commercial input was harmful**.

Cycle-02 Rolling Forecast & Lifecycle Handoff

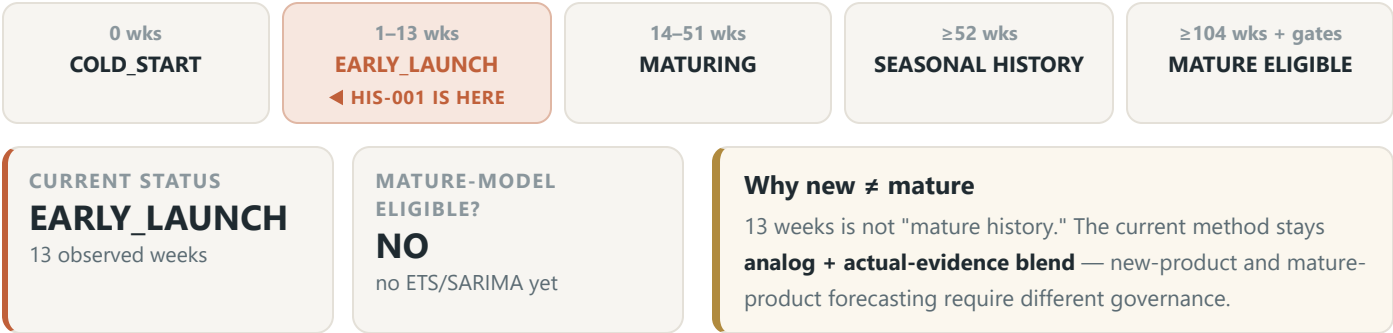
A · A GENUINE ROLLING FORECAST



September becomes **actualized**; **March 2028 enters as a newly-generated Month 18** (same frozen seasonal foundation, not copied). Learning is attenuated by lifecycle phase — near-term **1.00** · normalization **0.50** · second season **0.25**

SYNTHETIC ROLLING-FORECAST GOVERNANCE ASSUMPTIONS

B · LIFECYCLE HANDOFF



What This Project Demonstrates

CAPABILITIES DEMONSTRATED

FORECASTING

Reconstruction · Backtesting · Scenario planning · FVA

PLANNING & S&OE

13-week S&OE · Inventory/service risk · Exception mgmt · Governance

IBP / NEW PRODUCT

Consensus · Analog · Cold start · Initial buy · Allocation · Rolling forecast · Handoff

GOVERNANCE

- PUBLIC / SYNTHETIC / DERIVED — **no Arc'teryx internal data**
- Hidden truth = **evaluation only**
- Economics = **planning exposure proxies**, not profit
- Enterprise concepts demonstrated — no unsupported platform-experience claim

LIMITATIONS

- Synthetic HIS launch · single seeded path
- Simplified supplier constraints; weather caps are assumptions
- FVA affected by V0-centered simulation design
- 13 launch weeks insufficient for mature-model fitting

THE TRANSFORMATION

DemandIQ moved the question from "*What is likely to sell?*" to:

"What demand should we plan for, what inventory and service risk does that create, what action should the planner take, how should a new product be launched, and how should the plan learn as new evidence arrives?"

The value is in **planning decision quality and governance** — not predictive accuracy alone.